

## ACTIVITY 13 — FACILITATOR GUIDE

### Solution Selection Matrix, Benchmarking, and FMEA

| TYPE              | DURATION | DMAIC PHASE | MAPS TO |
|-------------------|----------|-------------|---------|
| Elective Activity | 60 min   | IMPROVE     | A5, K2  |

#### Purpose of this activity

Evaluate and de-risk candidate solutions before implementation (A5, K2).

Three Improve-phase tools from the original course. Score solutions against weighted criteria, benchmark against outstanding practice, and use FMEA to find how a proposed change could fail before it does.

#### What learners produce

A scored solution selection matrix ranked by weighted score, a benchmarking summary and an FMEA with RPN and actions on the highest scores.

#### Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- A laptop with Excel, Google Sheets or LibreOffice Calc
- The CSV files in this activity's data/ and templates/ folders
- Flipchart or A3 sheet and markers
- The course slides for the scenario and the framework

#### Set-up

Groups of 3-5. Distribute one worksheet per learner and one checklist per group. Every activity runs on the same continuous Contoso Service Desk scenario, so remind learners where this one sits in that story before starting.

#### Scenario / briefing material

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

## Files in this activity folder

### data/ — the mock data you analyse:

- candidate-solutions.csv — 5 rows (solution\_id, candidate\_solution, est\_cost, est\_effort\_days)

### templates/ — the worksheets you complete:

- fmea.csv (process\_step, failure\_mode, effect, cause, severity\_1\_10, occurrence\_1\_10, detection\_1\_10, RPN, action)
- solution-selection-matrix.csv (solution, impact\_w5, cost\_w3, ease\_w2, weighted\_score, rank)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

### How to run it — step by step

| # | Instruction to learners                                                                    | Facilitator note                                                                               |
|---|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 1 | Agree the selection criteria and weight each by importance.                                | Weights must be agreed BEFORE scoring, or the scoring is rationalised backwards.               |
| 2 | Score every candidate solution against the weighted criteria and rank the results.         | candidate-solutions.csv deliberately includes one over-engineered option — watch who picks it. |
| 3 | Benchmark your process against an internal or external reference and note the gap.         |                                                                                                |
| 4 | Build an FMEA: failure mode, effect, cause, then score Severity, Occurrence and Detection. |                                                                                                |
| 5 | Calculate RPN = S x O x D and address the highest-RPN failure modes first.                 |                                                                                                |

### Suggested timing

| Minutes | Phase                                   |
|---------|-----------------------------------------|
| 10 min  | Agree and weight the selection criteria |
| 15 min  | Score the candidate solutions           |
| 10 min  | Benchmark against a reference           |
| 15 min  | Build the FMEA                          |
| 10 min  | Calculate RPN and prioritise            |

**Total: 60 minutes**

### Discussion & decision prompts

1. Did the weighted score change the ranking versus your gut preference?
2. Which failure mode has the highest RPN, and is it the one you feared?
3. Is a high-severity, low-occurrence risk more dangerous than the reverse?

### Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what the data showed.
- Link the activity explicitly back to the IMPROVE phase of DMAIC.
- Remind learners this activity rehearses what the Case Study assessment will require.

Reflection questions:

- What change at your workplace failed in a way an FMEA would have caught?
- Who would you benchmark against, and would they talk to you?

### **Facilitator notes — what to watch for**

- Weights must be agreed BEFORE scoring, or the scoring is rationalised backwards.
- candidate-solutions.csv deliberately includes one over-engineered option — watch who picks it.

### **Success criterion**

Your matrix ranks solutions by weighted score, and every FMEA row has an RPN and an action for the highest scores.

### **Assessment link**

This activity builds the IMPROVE phase of DMAIC. All activities are in-class activities assessed indirectly through the open-book Case Study (CS), which reuses the same techniques on the same Contoso Service Desk scenario and the same figures (400 tickets, yield 76%, DPMO 50,000). Knowledge is assessed in the Written Assessment (WA-SAQ). Nothing is assessed that is not practised here or taught in the slides.